

Chapter 9: Linear Programming

Chapter overview

Linear programming optimizes a linear objective over a region determined by linear constraints. The chapter develops the geometry of half-planes and feasible polygons, the vertex theorem, the simplex method, minimization through duality, and mixed-constraint models through the two-phase method.

Original contexts

All worked numerical examples are original. Themes selectively use the San Antonio Spurs, Backstreet Boys, hiking, badminton, driving, travel, and R&B music. All business and scheduling contexts are fictional teaching models.

Exact arithmetic convention

Simplex tableaus are carried out with exact integers and fractions. Decimal rounding is postponed until an application explicitly requires it.

1 Systems of Linear Inequalities

Linear inequality

A linear inequality in two variables has the form

$$ax + by < c, \quad ax + by \leq c, \quad ax + by > c, \quad \text{or} \quad ax + by \geq c,$$

where a and b are not both zero.

The associated boundary is the line $ax + by = c$. A strict inequality excludes the boundary and uses a dashed line; a non-strict inequality includes the boundary and uses a solid line.

Why one test point determines a half-plane

Let $L(x, y) = ax + by - c$. Within either open half-plane, a line segment joining two points never crosses $L = 0$. The function L is continuous along the segment. Its sign cannot change without passing through zero, so all points in one half-plane have the same sign of L . Testing one point therefore identifies the entire solution side.

Graphing $2x - y \leq 4$

The boundary is

$$y = 2x - 4.$$

It is solid. Testing $(0, 0)$ gives $0 \leq 4$, so the side containing the origin is shaded. Solving for y gives

$$2x - y \leq 4 \iff y \geq 2x - 4,$$

because multiplying by -1 reverses the inequality.

Systems and convexity

A system is an intersection of half-planes. If $F = H_1 \cap \cdots \cap H_k$, then a point is feasible precisely when it belongs to every H_i .

Convexity of a feasible region

The feasible region of a system of linear inequalities is convex.

Proof

Let $\mathbf{u}, \mathbf{v}$ be feasible and $0 \leq t \leq 1$. For a constraint $\mathbf{a}^\top \mathbf{x} \leq b$,

$$\mathbf{a}^\top (t\mathbf{u} + (1-t)\mathbf{v}) = t\mathbf{a}^\top \mathbf{u} + (1-t)\mathbf{a}^\top \mathbf{v} \leq tb + (1-t)b = b.$$

Thus every convex combination is feasible. The argument applies to every constraint.

A five-vertex feasible polygon

For

$$x, y \geq 0, \quad x + y \leq 8, \quad 2x + y \leq 12, \quad x + 2y \leq 14,$$

the feasible vertices are

$$(0, 0), (6, 0), (4, 4), (2, 6), (0, 7).$$

For example, intersecting $x + y = 8$ and $2x + y = 12$ gives $x = 4, y = 4$. Intersecting $x + y = 8$ and $x + 2y = 14$ gives $x = 2, y = 6$.

Fictional hiking and badminton model

Let x be long hiking sessions and y badminton sessions. The inequalities above can represent total-session, recovery, and facility/travel limits. Nonnegativity is required because session counts cannot be negative.

Section 9.1 summary

Graph each boundary, identify the correct side, intersect all solution regions, and compute vertices from active boundary equations.

2 Two-Variable Linear Programming

Linear programming model

A linear programming problem maximizes or minimizes

$$f(\mathbf{x}) = \mathbf{c}^\top \mathbf{x}$$

subject to linear constraints. A feasible point satisfies all constraints; an optimal point has the best feasible objective value.

The level lines $c_1x + c_2y = k$ are parallel. Moving them in the improving direction gives a geometric picture of optimization.

Vertex principle

If a linear objective has a finite optimum over a nonempty polygonal feasible region, then at least one optimal point is a vertex.

Detailed proof

If an optimal point $\mathbf{x}^*$ is not a vertex, write it as a nontrivial convex combination

$$\mathbf{x}^* = t\mathbf{u} + (1-t)\mathbf{v}, \quad 0 < t < 1,$$

with distinct feasible $\mathbf{u}, \mathbf{v}$. Linearity gives

$$f(\mathbf{x}^*) = tf(\mathbf{u}) + (1-t)f(\mathbf{v}).$$

For a maximum, neither endpoint can exceed $f(\mathbf{x}^*)$. Their weighted average equals $f(\mathbf{x}^*)$, so both endpoints must also be optimal. Repeat on the face until an optimal vertex is reached. The minimum case follows by maximizing $-f$.

Fictional Spurs fan-night bundles

Maximize

$$P = 70x + 60y$$

subject to the five-vertex region from Section 9.1. The objective values are

(x, y)	$(0, 0)$	$(6, 0)$	$(4, 4)$	$(2, 6)$	$(0, 7)$
P	0	420	520	500	420.

Hence $P_{\max} = 520$ at $(4, 4)$.

Multiple optima and unboundedness

If an objective level line is parallel to an active feasible edge, every point on the touching edge is optimal. An unbounded feasible region may still have a finite minimum, but an objective improving along a recession direction is unbounded.

Why all points between two optimizers are optimal

If $\mathbf{u}$ and $\mathbf{v}$ are feasible and $f(\mathbf{u}) = f(\mathbf{v}) = M$, then for $0 \leq t \leq 1$,

$$f(t\mathbf{u} + (1-t)\mathbf{v}) = tM + (1-t)M = M.$$

Convexity makes the entire segment feasible.

Section 9.2 summary

Formulate the model, compute every feasible vertex, evaluate the objective, verify constraints, and interpret active constraints and slack.

3 The Simplex Method: Maximization

Standard form

A standard maximization problem is

$$\max z = \mathbf{c}^\top \mathbf{x}, \quad A\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \geq 0, \quad \mathbf{b} \geq 0.$$

Add slack variables to obtain equations.

Basic feasible solution

With m independent equations, choose m basic variables, set all nonbasic variables to zero, and solve. If every variable is nonnegative, the result is a basic feasible solution.

Entering-variable rule

Under the convention $z + \sum r_j x_j = z_0$, any nonbasic variable with $r_j < 0$ can initially increase the objective.

Proof

Solving for the objective gives

$$z = z_0 - \sum r_j x_j.$$

If $r_j < 0$ and x_j increases from zero, the contribution $-r_j x_j$ is positive.

Ratio test

When x_j enters, the maximum feasible increase is

$$\min_{a_{ij} > 0} \frac{b_i}{a_{ij}}.$$

Proof

A basic variable has the form $x_{B_i} = b_i - a_{ij}x_j$. For $a_{ij} > 0$, nonnegativity requires $x_j \leq b_i/a_{ij}$. The smallest such bound is reached first and identifies the departing variable.

Optimality criterion

If all reduced costs in a feasible tableau are nonnegative, the current solution is optimal.

Proof

From $z + \sum r_j x_j = z_0$ with $r_j, x_j \geq 0$,

$$z = z_0 - \sum r_j x_j \leq z_0.$$

The current solution achieves z_0 .

Complete two-pivot simplex calculation

Maximize

$$z = 3x_1 + 5x_2$$

subject to

$$x_1 \leq 4, \quad x_2 \leq 6, \quad 3x_1 + 2x_2 \leq 18, \quad x_1, x_2 \geq 0.$$

The initial tableau is

	x_1	x_2	s_1	s_2	s_3	b
R_1	1	0	1	0	0	4
R_2	0	1	0	1	0	6
R_3	3	2	0	0	1	18
R_z	-3	-5	0	0	0	0

x_2 enters. Ratios are 6 and 9, so s_2 leaves. Use

$$R_3 \leftarrow R_3 - 2R_2, \quad R_z \leftarrow R_z + 5R_2.$$

Then x_1 enters. Ratios are 4 and 2, so s_3 leaves. Use

$$R_3 \leftarrow \frac{1}{3}R_3, \quad R_1 \leftarrow R_1 - R_3, \quad R_z \leftarrow R_z + 3R_3.$$

The final tableau is

	x_1	x_2	s_1	s_2	s_3	b
R_1	0	0	1	$2/3$	$-1/3$	2
R_2	0	1	0	1	0	6
R_3	1	0	0	$-2/3$	$1/3$	2
R_z	0	0	0	3	1	36.

Thus $x_1 = 2$, $x_2 = 6$, and $z_{\max} = 36$.

Exceptional cases

No positive entry in an entering column proves unboundedness. A zero minimum ratio creates degeneracy. A zero reduced cost for a nonbasic variable in a final tableau signals an alternate optimum. Cycling can be prevented by Bland's rule.

Section 9.3 summary

Every pivot is a Gauss–Jordan basis change. The reduced-cost rule seeks improvement, the ratio test preserves feasibility, and the final nonnegative bottom row is an algebraic certificate of optimality.

4 Minimization and Duality

For

$$\min \mathbf{c}^\top \mathbf{x}, \quad A\mathbf{x} \geq \mathbf{b}, \quad \mathbf{x} \geq 0,$$

the dual is

$$\max \mathbf{b}^\top \mathbf{y}, \quad A^\top \mathbf{y} \leq \mathbf{c}, \quad \mathbf{y} \geq 0.$$

Weak duality

For every primal-feasible $\mathbf{x}$ and dual-feasible $\mathbf{y}$,

$$\mathbf{b}^\top \mathbf{y} \leq \mathbf{c}^\top \mathbf{x}.$$

Proof

Since $A\mathbf{x} \geq \mathbf{b}$ and $\mathbf{y} \geq 0$,

$$\mathbf{y}^\top A\mathbf{x} \geq \mathbf{y}^\top \mathbf{b}.$$

Since $A^\top \mathbf{y} \leq \mathbf{c}$ and $\mathbf{x} \geq 0$,

$$\mathbf{x}^\top A^\top \mathbf{y} \leq \mathbf{x}^\top \mathbf{c}.$$

The middle scalars are equal, yielding the result.

Strong duality

When a finite optimum exists, the primal and dual optimal values are equal.

Simplex-basis proof

At an optimal primal basis B , let $\mathbf{y}^\top = \mathbf{c}_B^\top B^{-1}$. Optimal reduced costs imply dual feasibility. The slack-column reduced costs imply $\mathbf{y} \geq 0$. Finally,

$$\mathbf{b}^\top \mathbf{y} = \mathbf{c}_B^\top B^{-1} \mathbf{b} = \mathbf{c}_B^\top \mathbf{x}_B.$$

The feasible primal and dual values coincide; weak duality makes both optimal.

Detailed minimization by the dual

Minimize

$$w = 6x_1 + 4x_2$$

subject to

$$x_1 + x_2 \geq 5, \quad 2x_1 + x_2 \geq 8, \quad x_1, x_2 \geq 0.$$

The dual is

$$\max z = 5y_1 + 8y_2$$

subject to

$$y_1 + 2y_2 \leq 6, \quad y_1 + y_2 \leq 4, \quad y_1, y_2 \geq 0.$$

The initial dual tableau is

	y_1	y_2	s_1	s_2	b
R_1	1	2	1	0	6
R_2	1	1	0	1	4
R_z	-5	-8	0	0	0.

First y_2 enters. The ratios are $6/2 = 3$ and $4/1 = 4$, so s_1 leaves. After

$$R_1 \leftarrow \frac{1}{2}R_1, \quad R_2 \leftarrow R_2 - R_1, \quad R_z \leftarrow R_z + 8R_1,$$

the tableau is

	y_1	y_2	s_1	s_2	b
R_1	1/2	1	1/2	0	3
R_2	1/2	0	-1/2	1	1
R_z	-1	0	4	0	24.

Then y_1 enters. Ratios are $3/(1/2) = 6$ and $1/(1/2) = 2$, so s_2 leaves. Use

$$R_2 \leftarrow 2R_2, \quad R_1 \leftarrow R_1 - \frac{1}{2}R_2, \quad R_z \leftarrow R_z + R_2.$$

The final tableau is

	y_1	y_2	s_1	s_2	b
R_1	0	1	1	-1	2
R_2	1	0	-1	2	2
R_z	0	0	3	2	26.

Thus $y_1 = y_2 = 2$ and $z = 26$. In the final dual tableau, the coefficients under the dual slack columns are 3 and 2. These recover the primal decision variables:

$$x_1 = 3, \quad x_2 = 2.$$

Verify primal feasibility:

$$x_1 + x_2 = 3 + 2 = 5, \quad 2x_1 + x_2 = 2(3) + 2 = 8.$$

The primal objective value is

$$w = 6(3) + 4(2) = 26.$$

Strong duality therefore gives

$$w_{\min} = z_{\max} = 26.$$

For complementary slackness, compute all four products. The primal slacks are

$$(x_1 + x_2) - 5 = 0, \quad (2x_1 + x_2) - 8 = 0,$$

and the dual slacks are

$$6 - (y_1 + 2y_2) = 6 - (2 + 4) = 0, \quad 4 - (y_1 + y_2) = 4 - (2 + 2) = 0.$$

Hence

$$\begin{aligned} y_1((x_1 + x_2) - 5) &= 2(0) = 0, & y_2((2x_1 + x_2) - 8) &= 2(0) = 0, \\ x_1(6 - (y_1 + 2y_2)) &= 3(0) = 0, & x_2(4 - (y_1 + y_2)) &= 2(0) = 0. \end{aligned}$$

Complementary slackness

At optimality,

$$x_j(c_j - (A^T \mathbf{y})_j) = 0, \quad y_i((A\mathbf{x})_i - b_i) = 0.$$

Proof

The duality gap decomposes as

$$\mathbf{c}^\top \mathbf{x} - \mathbf{b}^\top \mathbf{y} = \mathbf{x}^\top (\mathbf{c} - A^\top \mathbf{y}) + \mathbf{y}^\top (A\mathbf{x} - \mathbf{b}).$$

Every term is nonnegative. Strong duality makes the sum zero, so every product is zero.

Section 9.4 summary

The dual supplies bounds, certificates, shadow prices, and a computational route for minimization. Equality of primal and dual values certifies optimality.

5 Mixed Constraints and Two-Phase Simplex

Variable types

For $\leq$, add a slack variable. For $\geq$, subtract a surplus variable and add an artificial variable. Artificial variables create a temporary initial basis and must be removed.

Phase I feasibility problem

Minimize

$$w = a_1 + \cdots + a_k.$$

Because every $a_i \geq 0$, $w \geq 0$.

Phase I feasibility criterion

The original constraints are feasible if and only if the Phase I minimum is zero.

Proof

If $w = 0$, every nonnegative artificial variable is zero, leaving a feasible solution of the original equations. Conversely, any original feasible solution extends to a Phase I solution by setting all artificial variables to zero. If the Phase I minimum is positive, no original feasible solution exists.

Complete two-phase setup

Maximize

$$z = 5x_1 + 4x_2$$

subject to

$$x_1 + x_2 \leq 6, \quad 2x_1 + x_2 \geq 4, \quad x_1, x_2 \geq 0.$$

Write

$$\begin{aligned} x_1 + x_2 + s_1 &= 6, \\ 2x_1 + x_2 - s_2 + a_2 &= 4. \end{aligned}$$

The canonical Phase I tableau is

	x_1	x_2	s_1	s_2	a_2	b
R_1	1	1	1	0	0	6
R_2	2	1	0	-1	1	4
R_w	-2	-1	0	1	0	-4.

Phase I pivot. The most negative coefficient in the Phase I objective row is -2 , so x_1 enters. The eligible ratios are

$$\frac{6}{1} = 6, \quad \frac{4}{2} = 2.$$

The smaller ratio occurs in R_2 , so the artificial variable a_2 leaves. Apply

$$R_2 \leftarrow \frac{1}{2}R_2, \quad R_1 \leftarrow R_1 - R_2, \quad R_w \leftarrow R_w + 2R_2.$$

This gives

	x_1	x_2	s_1	s_2	a_2	b
R_1	0	1/2	1	1/2	-1/2	4
R_2	1	1/2	0	-1/2	1/2	2
R_w	0	0	0	0	1	0.

The Phase I objective value is $w = 0$, and a_2 is nonbasic. Therefore the original constraints are feasible.

Restore the original objective. Delete the artificial-variable column. Start with

$$z - 5x_1 - 4x_2 = 0.$$

Because x_1 is basic in R_2 , eliminate the x_1 coefficient from the objective row by applying

$$R_z \leftarrow R_z + 5R_2.$$

The initial Phase II tableau is

	x_1	x_2	s_1	s_2	b
R_1	0	1/2	1	1/2	4
R_2	1	1/2	0	-1/2	2
R_z	0	-3/2	0	-5/2	10.

There are several valid choices of a negative reduced cost. To illustrate that the simplex rule does not require choosing the most negative entry, choose x_2 first.

Phase II pivot 1. For the x_2 column, the ratios are

$$\frac{4}{1/2} = 8, \quad \frac{2}{1/2} = 4.$$

Thus x_2 enters and x_1 leaves. Apply

$$R_2 \leftarrow 2R_2, \quad R_1 \leftarrow R_1 - \frac{1}{2}R_2, \quad R_z \leftarrow R_z + \frac{3}{2}R_2.$$

The tableau becomes

	x_1	x_2	s_1	s_2	b
R_1	-1	0	1	1	2
R_2	2	1	0	-1	4
R_z	3	0	0	-4	16.

Phase II pivot 2. The only negative reduced cost is now -4 in the s_2 column. Only R_1 has a positive entry in that column, so s_2 enters and s_1 leaves. The pivot is already 1. Apply

$$R_2 \leftarrow R_2 + R_1, \quad R_z \leftarrow R_z + 4R_1.$$

This gives

	x_1	x_2	s_1	s_2	b
R_1	-1	0	1	1	2
R_2	1	1	1	0	6
R_z	-1	0	4	0	24.

Phase II pivot 3. The coefficient -1 under x_1 is negative, so x_1 enters. Only R_2 has a positive entry in the entering column, and the ratio is $6/1 = 6$. Thus x_2 leaves. Eliminate the x_1 entries using

$$R_1 \leftarrow R_1 + R_2, \quad R_z \leftarrow R_z + R_2.$$

The final tableau is

	x_1	x_2	s_1	s_2	b
R_1	0	1	2	1	8
R_2	1	1	1	0	6
R_z	0	1	5	0	30.

The nonbasic variables are $x_2 = s_1 = 0$. Therefore

$$x_1 = 6, \quad s_2 = 8,$$

and

$$z_{\max} = 5(6) + 4(0) = 30.$$

The surplus value verifies the second constraint:

$$2x_1 + x_2 - s_2 = 12 - 8 = 4.$$

The final objective row can be written as

$$z + x_2 + 5s_1 = 30,$$

so every feasible solution satisfies $z \leq 30$. This is an algebraic certificate of optimality.

Mixed minimization

Minimizing $w = \mathbf{c}^\top \mathbf{x}$ is equivalent to maximizing $z = -\mathbf{c}^\top \mathbf{x}$. Constraint conversion and Phase I are unchanged.

Transportation interpretation

Transportation models combine supply upper bounds, demand lower bounds, and nonnegative shipments. Specialized network algorithms are efficient, but two-phase simplex remains a valid general method.

Section 9.5 summary

Phase I answers whether a feasible starting basis exists. Phase II optimizes the original model. A positive Phase I minimum is an exact certificate of infeasibility.

Chapter terminology

Term	Meaning
Linear inequality	A comparison involving a linear expression
Feasible region	Intersection of all constraint regions
Vertex	Corner point of a polyhedral feasible set
Objective function	Quantity to maximize or minimize
Basic feasible solution	Nonnegative solution associated with a basis
Slack	Unused capacity in a $\leq$ constraint
Surplus	Excess above a $\geq$ requirement
Reduced cost	Objective-row coefficient in canonical form
Dual problem	Transposed companion optimization model
Shadow price	Marginal value associated with a constraint
Artificial variable	Temporary variable used to construct a Phase I basis
Degeneracy	Basic feasible solution with a zero basic variable

Closing perspective

Linear programming is a unified theory: convex geometry explains vertices, matrix algebra represents basis changes, simplex tableaus implement those changes, and duality proves that a candidate solution is globally optimal.